

## SECTION 2: Engineering Economics (CE 655)

### Paper 1: 2082 Kartik

Attempt all questions.

1. [Q1]
  - (a) Define Engineering Economics. Explain its role in decision-making processes for engineering projects.
  - (b) Deposit an equal annual amount  $A$  for 12 years ( $t=1 \dots 12$ ) to withdraw Rs. 3,000 per year for 5 years beginning at the end of the 14<sup>th</sup> year ( $t=14, 15, 16, 17, 18$ ). Interest rate  $i = 9\%$  compounded quarterly. Calculate  $A$ .
2. [Q2]
  - (a) A loan of Rs. 100,000 is repaid in equal monthly payments of Rs. 5,500 over 24 months. If average monthly inflation rate  $f = 0.75\%$  per month, calculate the inflation-free monthly interest rate ( $i'$ ).
  - (b) An annual deposit  $A = \text{Rs. } 1,500$  is made at  $i = 7\%$  per year. How many years  $n$  will it take to reach a target future sum  $F = \text{Rs. } 25,000$ ?
3. [Q3]
  - (a) Monthly deposit  $A_1 = \text{Rs. } 5,000$  is made for 5 years ( $n = 60 \text{ months}$ ) at  $i = 12\%$  per year compounded monthly. Find the maximum fixed monthly amount  $W$  that can be withdrawn indefinitely after 5 years while keeping the principal balance intact.
  - (b) Differentiate between Nominal Interest Rate and Effective Interest Rate.
4. [Q4] An asset has Initial Cost  $B = \text{Rs. } 80,000$ , Salvage Value  $S = \text{Rs. } 10,000$ , and Useful Life  $N = 5 \text{ years}$ . Calculate annual depreciation  $D_t$  and book value  $BV_t$  for each year using:
  - (a) Double Declining Balance (DDB)
  - (b) Sum-of-the-Years'-Digits (SOYD)
  - (c) Sinking Fund Method (using  $i = 10\%$ )

### Paper 2: 2082 Jestha

Attempt all questions.

1. [Q1] Explain Simple Interest vs. Compound Interest. Draw cash flow diagrams for single payment, uniform series, and gradient series.
2. [Q2] Define Net Present Value (NPV), Internal Rate of Return (IRR), and Benefit-Cost (B/C) ratio. Compare mutually exclusive alternatives using NPV and IRR analysis.
3. [Q3] What is payback period? Differentiate between simple payback period and discounted payback period with advantages and limitations.
4. [Q4] Explain replacement analysis. Define defender asset, challenger asset, and sunk cost with examples.

5. [Q5] Explain the concept of inflation and purchasing power. Differentiate between real cash flows and actual (nominal) cash flows.
6. [Q6] Write short notes on:
  - (a) Capital recovery factor
  - (b) Sensitivity analysis in economic decision making

### **Paper 3: 2081 Ashwin**

**Attempt all questions.**

1. [Q1] Explain cost concepts in engineering decision making: Fixed cost, Variable cost, Marginal cost, and Sunk cost.
2. [Q2] Describe the Annual Worth (AW) method and Future Worth (FW) method for evaluating investment projects.
3. [Q3] What is Break-Even Analysis? Derive the break-even point formula for quantity produced and chart out fixed, variable, and total revenue curves.
4. [Q4] Explain straight-line depreciation, declining balance depreciation, and unit-of-production depreciation methods with formulas.
5. [Q5] Discuss project financing sources: Equity financing vs. Debt financing. Explain Capital Budgeting process.
6. [Q6] Write short notes on:
  - (a) Minimum Attractive Rate of Return (MARR)
  - (b) Risk assessment and decision trees

### **Paper 4: 2079 Chaitra**

**Attempt all questions.**

1. [Q1] Explain the basic economic principles used in engineering project evaluation. Define time value of money.
2. [Q2] Describe continuous compounding interest calculations. Derive the relation  $S_i = e^{rt} - 1$ .
3. [Q3] Explain Benefit-Cost Ratio Analysis (B/C ratio) for public sector projects. Differentiate between conventional and modified B/C ratios.
4. [Q4] Define tax laws, taxable income, corporate income tax, and corporate cash flow after taxes (CFAT).
5. [Q5] Explain risk analysis in project evaluation using Monte Carlo simulation and scenario analysis.
6. [Q6] Write short notes on:
  - (a) Economic life of an asset
  - (b) Life-cycle cost analysis (LCCA)